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If you want to learn computer programming then contact with me

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MTH401 MIDTERM PAST PAPER SHARED BY STUDENT IN 2017

My Paper today!

MCQs were eventually easy and they weren't that much hard to solve. Most of the MCQs were from chapter#3-12

2 marks questions:

1. Variations of parameters to find the value of w_1
2. Show independence b/w $p(x)$ and $g(x)$

3 marks questions:

1. To find IF of a differential equation given
2. Showing the equation to prove coefficients (given from chapter 16)

5 marks questions:

1. First order linear equation to be solved DE was given.
2. Finding Annihilator operator approach of a given DE



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Q1. $Y = emx$ then construct an auxiliary equation for

$$d^2y/dx^2 + dy/dx + 2y = 0$$

Q2. The general solution is the combination of which two solution ???

Q3. $m^2 + 8m + 16$ is the solution of $d^2y/dx^2 + 8dy/dx + 16 = 0$ then find the general solution

Q4. Show the following equation in the form of $h(x) = g(y)$

$$d^2y/dx^2 - 2 dy/dx + 5y = 2 \sin y$$

Q5. By using the newton law show that

$$T(t_1 - t_m)/T(t_2 - t_m) = e^{k(t_1 - t_2)}$$

When $t(0) = t_0$ is initial condition

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Paper 1:-

Question No 1: Define cycle of a vibrating body (2)

Question No 2: Construct the auxiliary equation of the following differential equation

$$3 d^2/dx^2 - 4 dy/dx = 0$$

Question No 3: Solve the following initial value problem

$$(\text{Integral sign}) dT / T - T_m = (\text{integral sign}) K(dt), T(0) = T_0$$

Question No 4: Factorize the differential equation $D^4 - 8D^2 + 16$ if possible

Question No 5: Find the wronsktin of the differential equation

$$Y'''' - 2y'' - 18y = 6 + 4 - t \text{ using variation of parameter and the root of the auxiliary equation is } m_1 = -3, m_2 = -1, m_3 = 6?$$

Question No 6: A radioactive isotope has a half-life of 16 days we have 30 g at the end of 30 days, how much radio isotope was initially present?

The solution of the IVP (initial value Problem) is given by $A(T) = A_0 e^{kt}$, where $K = \ln 2 / 16$



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Paper 2:-

Question No. 1) Define super position principle of particular solution non Homogeneous Linear deferential equation?

Question No. 2) If a mass weighing 15lb stretches a spring 3ft then find K using hook's law? vuzs.net

Question No. 3) what is an auxiliary equation of a Homogeneous differential equation .also give its two examples.

Question No. 4) what is the general solution of differential equation?

Question No. 5) Take a differential equation

2

2 0

$\frac{dy}{dx} + a y = b e^{cy}$

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$\frac{dy}{dx} + a y = b e^{cy}$

$\frac{dy}{dx} + a y = b e^{cy}$ we take its exponential solution

$y = e^{mx}$, by putting the above solution we get equation $(a^2 + b^2)m^2 + 2abm + c = 0$. What will be its general solution if roots are distinct real roots?

Question No. 6) Find the complementary solution for the equation $(D^2 + 3D + 2)y = x + e^x$

Paper 3:-

MTH401 Q.No.1) Define super position principle of particular solution non Homogeneous Linear differential equation?

MTH401 Q.NO.2) If a mass weighing 15lb stretches a spring 3ft then find K using hook's law? vuzs

MTH401 Q.No.3) what is an auxiliary equation of a Homogeneous differential equation .also give its two examples.

Paper 4:-

Q.1. write all the forms of $g(x)$ in the method of undetermined coefficients where $g(x)$ is right hand side of Non-homogeneous linear differential equation ?

Q. 2. Deduce the special use of logistic Equation "Epidemic Spread"?

Q. 3. what will be the auxiliary Equation of 2nd order Homogeneous Linear Differential equation by taking solution of exponential form $y'' - 2y' + y = e^t/t^2 + 1$

Q. 4. find the auxiliary equation for the differential equation $y''' - 2y'' - 21y' - 18y = 3 + 4e^{-t}$

Q. 5. Define general linear equation of order n ?



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My today's paper of 11 o'clock:

paper was of 40 marks:

20 mcq's ,2 questions of 2 marks,2 questions of 3 marks,2 questions of 5 marks.

some **mcq's** was from past papers and some was new.

1-If half time(T) of radioactive isotope is 5,then find k. (2 marks)

2-Write equation of free undamped motion.(2 marks)

3-The equation was given,and it was to tell is it linear dependent or independent? (3 marks)

4-One was from find particular integral of the equation.(3 marks)

5-In this question,it was to write the annihilator operator and general equation.(5 marks)

6-Deduce the special use of logistic Equation "Epidemic Spread"?(5 marks)

ALL THE BEST TO ALL THE STUDENTS...!



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My today paper of Mth401:

20 mcqs

Question):21

solve the logical equation

$$dp/dt=p(a_{bp})$$

Question):22

Differential equation $d^3y/dx^3=\tan x$ write down the first column of w_1 ?

Question):23

Take the derivative of the given equation?

Question):24

find the auxiliary equation for the differential equation of

$$y^2+9y=3\tan(3t)$$

Question):25

find homogenous equation is given

bas itna yad ha paper

Best of luck All



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Q3. $m^2 + 8m + 16$ is the solution of $d^2y/dx^2 + 8dy/dx + 16 = 0$ then find the general solution

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